Exercise 10

This exercise takes a break from new material and revisits the suitability analysis lab most students did in GSP 531. A key part of that lab was using ModelBuilder in ArcGIS Pro. This time you will do it using Python. The "background" information from the GSP 531 lab is repeated here (with minor changes near the end) as a refresher to motivate this exercise. It is also important to point out that an analysis done using open source tools is even more accessible to others since they do not need to pay for software to run your analysis.

Background

Suitability Analysis

Suitability analysis leverages the GIS to find appropriate locations for an activity based on multiple criteria that are often in opposition to one another. If I want to locate a new Jack in the Box restaurant, I would want to locate it along a major roadway, ideally at a busy intersection and far from existing Jack in the Box locations. You might be thinking, "should it also be far from competitors?"; the answer is "no" since any rational consumer would choose Jack in the Box over McDonalds or Burger King ;-). Read the following website for more background on suitability analysis
https://atlas.co/glossary/suitability-analysis

Sensitivity Analysis

Now that you understand the basics of suitability analysis, a cynical student might say, "can't you just pick any criteria you want to get the outcome you want?" The answer is, "don't be so cynical, but you're right." Sensitivity analysis addresses this weakness by allowing you to tweak your assumptions to see which of them has the biggest impact on your final results. Read the following website for more background on sensitivity analysis.
https://www.statisticshowto.com/sensitivity-analysis/

Reproducibility and Replicability

There is a movement in the sciences for "reproducibility" and "replicability." This is also important in industry and government. Here are a couple of definitions:

- Reproducibility: A study is reproducible if you can take the original data and the computer code used to analyze the data and reproduce all of the numerical findings from the study. This may initially sound like a trivial task but experience has shown that it's not always easy to achieve this seemingly minimal standard.
- Replicability: This is the act of repeating an entire study based on a description of the study; i.e., independently of the original investigator without the use of their data, their programming code, etc. (but generally using the same methods).

Read this document (which is the source of the definitions above). The document is geared toward academic research and programming. However, the core ideas are relevant to working in any organization or with any software.
https://www.amstat.org/asa/files/pdfs/POL-ReproducibleResearchRecommendations.pdf

These concepts largely revolve around sharing your ideas with others, but they also help when working by yourself. How many times during this class when something when wrong did you think to yourself, "did I click that button or not", "what were the settings I passed into that tool", "how did I get this layer", etc. GIS processes are complex, and it is easy to lose track of what you did and where you are going.

What about the case where you thought, "what would happen if I did this a little differently." When there is unlimited time, you could just redo the whole exercise tweaking different settings each time to see what happens. Since there is not unlimited time, those "what if" questions often go unanswered. Now think about an important project at work. You just spent two weeks working out a complex analysis in the GIS and are presenting the results to your boss. It is not unreasonable that your boss will say, "those results are great, but what would happen if we double that input" or "that's a great idea, can you repeat that analysis on ten more areas." Repeating the analysis from scratch is likely a daunting task.

Yet another case is where you develop a workflow that you want to make easy for someone else to run. Maybe it is a process that staff needs to implement frequently; the people know the basics of ArcGIS Pro, but do not have the expertise of someone who has completed GSP ~~331/531~~ 538. What if you could ~~add a tool to the toolbox~~ create a Python script that encapsulates all the steps into a single ~~button~~ function? That would certainly make the other person's life easier and minimize the potential for errors if the person had to do all the steps one-by-one.

~~ArcGIS Pro provides a tool called Model Builder that~~ Python can address these scenarios and others. ~~The tool~~ A Python script allows you to chain the steps of your workflow together ~~in a graphical interface, and then run the entire workflow (i.e., "model") with a single click~~. Once the model is built, you can finetune just those parts of the model that need fixing without having to manually click through all the steps. You can make copies of the model to run "what if" scenarios. You can hand the model to someone else instead of explaining the minutia of each step needed to get to the result.

Arizona Wind Power Suitability Analysis[1]

Global warming and volatile oil prices make fossil fuel alternatives particularly appealing. In this activity you will identify the top locations in the state of Arizona for the installation of wind power generators.

Location criteria are as follows:

- Wind
    - Annual average wind resource potential at a 50 meter height. The shapefile was generated from a raster dataset with a 200 meter resolution. Areas are rated from 1 to 7, with 7 having the highest energy potential. The class 1 areas were removed to reduce the size of the file for the class project. The wind power resource estimates were produced by AWS TrueWind using their MesoMap system and historical weather data under contract to Wind Powering America/NREL. https://openei.org/doe-opendata/dataset/nrel-gis-data-arizona-high-resolution-wind-resource
    - az_wind_above_class1\arizona_50mwind.shp
    - Sites must have a wind power class code (GRIDCODE in the attribute table) of 3 or greater to ensure there is enough wind.
- Roads
    - Primary and secondary roads produced by the US Census Bureau TIGER/Line program. https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html
    - tl_2019_04_prisecroads\tl_2019_04_prisecroads.shp
    - Sites must be within 7.5 km of a major road to provide access for maintenance and construction.

[1] Inspired by https://frew.eri.ucsb.edu/private/ESM263/assignment/3/HW3.html

- Airports
    - U.S. National Atlas Airports represents airports in the United States, Puerto Rico, and U.S. Virgin Islands with airport passenger enplanements of greater than or equal to 25 passengers per year. It is maintained by Esri. https://www.arcgis.com/home/item.html?id=900d50de880644cdb90c4cab966d0e94
    - USA_Airports\USA_Airports.shp
    - Sites cannot be within 7.5 km of an airport due to accident potential and radar interference.
- Urban
    - Adjusted Urbanized Areas for Arizona as derived from the 2010 Census and smoothed by the Arizona Department of Transportation and approved by the Federal Highway Administration. An Urbanized Area is a Census-designated urban area having 50,000 or more residents (USDOT uses the acronym "UZA" to refer to Urbanized Areas, while Census uses “UA” to refer to the same type of area). These data include both Urban Areas, as described above, and Urban Clusters. An Urban Cluster is a Census-designated urban area having between 2,500 and 49,999 residents. https://azgeo.maps.arcgis.com/home/item.html?id=024d232820914dea89e96040085ca7f0
    - UZA_2013-shp\UZA_2013.shp
    - Sites cannot be within 1 km of an existing urban area.
- Public
    - Surface Management Agency (SMA) provides land ownership for the State of Arizona. Changes in federal agencies are done through the Bureau of Land Management (BLM). BLM and State of Arizona surface management is updated approximately every six months in a cooperative effort with the Arizona State Land Department (ASLD). https://azgeo.maps.arcgis.com/home/item.html?id=116e7b55ddd143b6950d708a4798d760
    - Land categories
        - Bankhead-Jones Act: Federally owned lands for rehabilitation
        - BLM: Bureau of Land Management
        - BR: Bureau of Reclamation
        - County: county owned lands
        - IBWC: International Boundary and Water Commission (US-Mexico water border)
        - Indian Lands
        - Local and State Parks
        - Military
        - NPS: National Parks Service
        - Other
        - Private: privately owned lands
        - State: state owned lands

        - State Wildlife Area
        - USFWS: US Fish and Wildlife Service
    - Arizona_Surface_Management_Agencies__BLM_-shp\Arizona_Surface_Management_Agencies__BLM_.shp
    - Sites cannot be on state or federal lands

Analysis steps

1. Open VS Code and create a new notebook in the `GSP538/exercises/` directory.

2. Read in the counties dataset from `GSP538/exercises/data/` and the five datasets introduced above from `GSP538/exercises/data/suitability_data` as geodataframes.
    a. Project each dataset to <u>NAD 1983 UTM Zone 12N</u>.
    b. Give the geodataframes meaningful names to help keep track of them.
    c. Print the CRS for one of the geodataframes to confirm you got the correct projection.

3. Create an interactive map of the data using the `explore` method. The following syntax allows you to include multiple layers on one map and places a layers button in the upper-right corner of the map allowing you turn layers on and off.

```
import folium
az_map = my_gdf1.explore(color='red', name='name1') # add layer 1
my_gdf2.explore(m=az_map, color='green', name='name2') # add layer 2
folium.LayerControl().add_to(az_map) # add the layers button to map
az_map # print the map
```

   <u>Warning</u>: the public and wind geodataframes are very large and slow to render so do not include them on the map at the same time.

4. [Intentionally blank to maintain the GSP 531 numbering]

5. Prepare each of the five data files and conduct the analysis. Hint: give the outputs short meaningful names that include the step number (e.g., roads_5b).

   Below is the syntax to create and then dissolve a buffer. The nuance is that the buffer method returns a geoseries, but the dissolve method only works on geodataframes. Also, notice that you could collapse these three lines into one line.

```
my_gdf_buff = my_gdf.buffer(1000)     # buffer each object
my_gdf_buff = my_gdf_buff.to_frame() # convert to geodataframe
my_gdf_buff = my_gdf_buff.dissolve() # dissolve into one object
```

   a. Wind: Create a new geodataframe with just the features with GRIDCODE greater than or equal to 3.
   b. Roads: Create a new geodataframe with a buffer of 7.5 km. Hint: Be sure to dissolve the buffers.
   c. Airports
      i. Create a new geodataframe with just the features with STATE equal to AZ.
      ii. Buffer the Arizona airports created in the previous step by 7.5 km. Hint: Be sure to dissolve the buffers.
   d. Urban: Create a new geodataframe with a buffer of 1 km. Hint: Be sure to dissolve the buffers.
   e. Public: Create a new geodataframe with just the features with CATEGORY equal to Private.

6. Identify areas that meet all of the criteria.
   a. Use the overlay function to get the intersection of Prepared Wind (output from 5a) and Prepared Roads (output from 5b). Note: this is slow.
   b. Use the overlay function to remove areas near airports (i.e., output from 5c) from the output from 6a. Hint: Esri calls this “erase”, geopandas calls this “difference.”
   c. Use the overlay function to remove areas near cities (i.e., output from 5d) from the output from 6b.
   d. Use the overlay function to get the intersection of Prepared Public (output from 5e) and the output from 6c.

7. Prepare your final set of areas for analysis.
   a. Note: you should have 1368 features in the output from 6d.
   b. Add county information to the final set of areas (output from 6d) using the overlay function.
   c. Remove unnecessary fields that have accumulated from the earlier processing steps so that only the CATEGORY, GRIDCODE and NAME attributes are left. Hint: do not forget to also include the geometry column or your geodataframe will turn into a regular pandas dataframe.
   d. Add a new column to the dataframe from 7c containing the area in acres of each feature.

8. Use the groupby function to sum the eligible acres by county (NAME) and wind class (GRIDCODE). Hint: do one groupby with both GRIDCODE and NAME; the resulting dataframe should have 43 rows. Also, compute the overall sum of this table to get the total acres for the whole state.

9. Create a map of the areas from your suitability analysis. It should just have counties and the 7d output (i.e., the final areas). Increase the size of the map and turn off the map borders. Make the county outlines visible. Remove the outlines on the final areas and use a fill color based on the wind class. Be sure that the colors of the selected areas and the color of the counties go well together. Note: Use a continuous color scheme for the class values, with darker colors showing higher wind class; the categories should just be 3, 4, 5 and 6. Move the legend off the map area.

10. Conduct a sensitivity analysis.
    a. Change one of the location suitability criteria (among Roads, Airports, Urban and Public). This implies changing the distance band (Roads, Airports, Urban) or the categories (Public). Rerun the suitability analysis.
    b. Reset the location suitability criteria to the baseline case and make a different change. Rerun the suitability analysis.

Deliverables

The pythonic way to conduct the suitability and sensitivity analysis would be to create a single function that runs the entire suitability analysis with parameters for road buffer distance, airport buffer distance, urban buffer distance and land categories. That way you could simply enter parameter values to run all kinds of scenarios. You are encouraged to use this approach but are not required to since this class does not require writing functions.

a. Organize and clean up your notebook
    i. Delete all unnecessary code and organize your cells. Include some basic documentation of your code.
    ii. Insert a Markdown cell above your code cells that contains the heading: Analysis
    iii. Insert a Markdown cell below your code cells that contains the heading: Results
    iv. Move the cell that produces the map of results into the Results section.
    v. The only cells in the Analysis section that print any output are the ones displaying the CRS (step 2) and the interactive map (step 3).
    vi. Insert four more Markdown cells in the Results section that contain subheadings (i.e., more #s): Baseline, Sensitivity Analysis 1, Sensitivity Analysis 2 and Discussion

b. Baseline
    i. Reset all the cells in the analysis area to the baseline values (i.e., before making sensitivity analysis changes) and rerun the analysis.
    ii. In the Baseline subsection, print the table of suitable acres. Note: this should only contain the code to print the table, not to build the table.
    iii. Print the total acres for the state.
    iv. Move your results map code into this section and run it to produce the map.

c. Sensitivity Analysis 1
    i. In the Analysis section, make one sensitivity analysis change and rerun the analysis.
    ii. In the results subsection, print the table of suitable acres.
    iii. Print the total acres for the state.
    iv. Copy and paste the map code into this subsection and produce the map.
    v. Insert a Markdown cell below the subsection heading. Briefly state the change you made, why you made the change and whether you expected that change to increase or decrease the suitable area.

d. Sensitivity Analysis 2
    i. Reset all the settings in the Analysis section to the baseline values (i.e., before making sensitivity analysis changes), make one sensitivity analysis change and rerun the analysis.
    ii. In the results subsection, print the table of suitable acres.
    iii. Print the total acres for the state.
    iv. Copy and paste the map code into this subsection and produce the map.
    v. Insert a Markdown cell below the subsection heading. Briefly state the change you made, why you made the change and whether you expected that change to increase or decrease the suitable area.

e. Discussion
    i. In a Markdown cell, write at least two paragraphs summarizing the results of your analysis. Note: this is not a discussion of the GIS steps you used to get to the maps. The discussion should include at a minimum: comments about the locations/distribution of feasible wind sites in Arizona, comments about information

from the suitable acres table, and an analysis of the differences between your suitability results and sensitivity results. You can optionally discuss differences between conducting this analysis in Python versus ModelBuilder.